Problem Set 5

QTM 220

Due 3 November at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

Gerber, Green, and Larimer Michigan Voter Experiment

Prior to the August 2006 primary election in Michigan, political scientists Gerber, Green, and Larimer conducted a large field experiment in which different mailers were sent to registered voters across Michigan. Because whether or not someone votes (though not who they vote for) is public record, the researchers then accessed the public voting information to measure their outcome– whether an individual voted in the August 2006 primary. The different treatments are as follows:

- Civic Duty: do your civic duty and vote!
- Hawthorne: you are being studied + do your civic duty and vote!
- Self: who votes is public information, here is your recent voting record + do your civic duty and vote!
- Neighbors: what if your neighbors knew whether you voted? here is your recent voting record + your neighbors' recent voting record + do your civic duty and vote!
- Control: no mailer

Households were randomized to receive a mailer (or not) and which type, but voting records are by individual.

The attached data from their experiment has the following variables:

- hh_id : the household id number
- hh_size : number of adults in household
- sex : individual's sex according to voter record

- yob: individual's year of birth
- g2000 : whether or not the individual voted in the general election in 2000
- g2002 : whether or not the individual voted in the general election in 2002
- g2004 : whether or not the individual voted in the general election in 2004
- p2000 : whether or not the individual voted in the primary election in 2000
- p2002 : whether or not the individual voted in the primary election in 2002
- p2004 : whether or not the individual voted in the primary election in 2004
- treatment : the mailer that the household received (or Control for no mailer)
- voted : whether the individual voted in the 2006 primary (outcome)

(a) Create a balance table showing the similarity of individuals assigned to treated and control across sex, age, household size, and turnout in previous elections.

(b) Create a balance table showing the similarity of individuals assigned to each level of treatment and control across sex, age, household size, and turnout in previous elections. Be sure to note how many individuals are in each level of treatment.

(c) Calculate the subsample 2006 primary turnout within each level of treatment (and control).

(d) Let's define the following treatment effects of interest: $\hat{\tau}(civic) = \hat{\mu}(civic) - \hat{\mu}(control)$, $\hat{\tau}(hawthorne) = \hat{\mu}(hawthorne) - \hat{\mu}(civic)$, $\hat{\tau}(self) = \hat{\mu}(self) - \hat{\mu}(hawthorne)$, $\hat{\tau}(neighbor) = \hat{\mu}(neighbor) - \hat{\mu}(self)$. Calculate each of these and interpret each in plain language.

(e) Create a bootstrap estimated sampling distribution for each of your τs and a 95% confidence interval for each.

(f) Create four dummy variables– one for each treatment– where an observation is coded as 1 if the household received that treatment and 0 otherwise (other treatment or control). Run an additive linear regression of 2006 primary vote on the four treatment variables. Report your results in a table and provide plain language interpretations of your estimates.

(g) Run an additive linear regression of 2006 primary vote on the four treatment variables and the pre-treatment turnout variables (i.e. general election voting in 2002 and 2000, primary election voting in 2004, 2002, and 2000). Report your results in a table (one table for both models is fine) and compare to your results in part (f).

Women's Health Initiative

The data based on the CEE+MPA arm of the WHI trial can be found in whidata.csv and contains the following variables:

- treat : binary indicating whether the participant received CEE+MPA or placebo
- age : the age group that the participant is in at the beginning of the trial
- breastcancer : binary indicating whether the participant developed invasive breast cancer during the trial

(a) Create a balance table showing the number of participants assigned to CEE+MPA and placebo by age group. Why do randomized experiments report balance tables? What do you think of the balance on age in the WHI?

(b) Estimate the overall treatment effect (difference in proportions) of CEE+MPA on breast cancer and interpret your result.

(c) Create a bootstrap estimated sampling distribution of your treatment effect and report your 95% confidence interval. Interpret your confidence interval.

(d) Estimate the treatment effect of CEE+MPA on breast cancer within each age group.

(e) Create a bootstrap estimated sampling distribution for the within-age-group treatment effects and report your 95% confidence interval for each age group. Interpret.

(f) Using the 50-59 year olds as the reference category, run an additive linear regression with breast cancer as your outcome and treatment, 60-69 year olds, and 70-79 year olds as your regressors. Report your results in a table and interpret.

(g) Using the 50-59 year olds as the reference category, run an interactive linear regression with breast cancer as your outcome and treatment, 60-69 year olds, 70-79 year olds, 60-69 year olds * treatment, and 70-79 year olds * treatment as your regressors. Report your results in a table (you can use one table for both models) and interpret.

(h) Write the equations for each or your models using β_is as your parameters. Fill in the following table using with the βs from your models (not numbers, for example $\beta_0 + \beta_1$.) Let $BC(T = 1) - BC(T = 0)$ be the difference in the proportion of those who developed invasive breast cancer that received treatment and control. Then fill in the table using the estimates from your models.

(i) Compare the results from your models to the $\hat{ATE}$s by age group that you recovered in part (d).

Additive Regression

	Age Group	Prop. Breast Cancer T=1	Prop. Breast Cancer T=0	$BC(T=1) - BC(T=0)$
βs	50-59			
βs	60-69			
βs	70-79			
numbers	50-59			
numbers	60-69			
numbers	70-79			

Interactive Regression

	Age Group	Prop. Breast Cancer T=1	Prop. Breast Cancer T=0	$BC(T=1) - BC(T=0)$
βs	50-59			
βs	60-69			
βs	70-79			
numbers	50-59			
numbers	60-69			
numbers	70-79			

Dogs

The Giant Database of Dogs, a crowd-sourced dataset organized by Prof. Leanne Powner, includes the following variables:

- Obs – unique observation number for case identification
- Name – dog's name as provided by the owner (excess text removed)
- Gender – {male, female} closed choice
- Fixed – "yes" if spayed or neutered, as appropriate; "no" if not
- Color – respondents selected one from {light brown, dark brown, black, white, reddish, yellow/blond, no primary color, other}. Other was not an open-ended response.
- Heritage – 1 = single breed, 2 = designer/deliberate mix, 3 = mixed/unknown

Use dogs.csv for the following.

(a) Recode gender and fixed into numerical binary variables. Regress dog's weight on gender and fixed. Report your results in a table.

(b) Create a scatter plot with weight on the y-axis, fixed on the x-axis, and use different colors for male and female dogs (include a legend).

(c) Regress dog's weight on gender, fixed, and gender*fixed. Report your results in a table (one table for both models is fine).

(d) Write the equations for each or your models using β_is as your parameters. Fill in the following table with the βs from your models (not numbers, for example $\beta_0 + \beta_1$.)

Weight: Additive Model βs

	Male	Female
Fixed		
Not Fixed		

Weight: Interactive Model βs

	Male	Female
Fixed		
Not Fixed		

(e) Fill in the table using the estimates from your models.

Weight: Additive Model Estimates

	Male	Female
Fixed		
Not Fixed		

Weight: Interactive Model Estimates

	Male	Female
Fixed		
Not Fixed		

(g) Calculate the subsample mean weights among fixed females, unfixed females, fixed males, and unfixed males. Compare your results to your additive and interactive tables of predicted weights.